

HUNG VINH TRAN

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SUMMARY

PhD candidate at UQ working on efficient machine learning models, with 4 years of prior industry experience at Cinnamon AI building production document-AI pipelines for clients in Japan and Taiwan. Comfortable across the full lifecycle: model research, PyTorch implementation, FastAPI serving, Docker, CI/CD and cloud deployment. Recently, I am working on both classical and LLM models with results that map directly to deployment cost – 90% smaller model memory footprint, 75% shorter training runs, and 93–95% of full accuracy from 1% of the training data.

EDUCATION

The University of Queensland

Doctor of Philosophy in Computer Science

Jul. 2023 – Present
Brisbane, QLD, Australia

- *Research Topic:* Scalable and Lightweight On-Device Recommender Systems.
- *Supervisor:* Assoc. Prof. Rocky Tong Chen (primary) & Prof. Hongzhi Yin

Ho Chi Minh City University of Science (HCMUS)

Bachelor of Science in Information Technology (Honors) | GPA: 8.68/10 – Top 5%

Sep. 2017 – Sep. 2021
Ho Chi Minh City, Vietnam

EMPLOYMENT HISTORY

Cinnamon AI, <https://cinnamon.ai>

Project Technical Leader & AI Engineer

Jul. 2021 – Jun. 2023
Ho Chi Minh City, Vietnam

- Led 3 document-AI projects end to end – PoC, model development, integration, production hand-over – with a 10-person team spanning Vietnam, Japan and Taiwan, covering printed and handwritten form types.
- Owned the technical relationship with client stakeholders and project managers: translated business requirements into accuracy, latency and hardware targets, then held the delivery to them.
- Built a custom LayoutLM-based clustering model for entity grouping and introduced Optuna-driven hyperparameter search to make results reproducible across noisy client datasets.
- Designed augmentation strategies that absorbed OCR and annotation noise, lifting model performance 5–10% without additional labelling spend.
- Shipped system running in production with client K-Line: kline.co.jp/en/news/drybulk/drybulk-20231023.html

Cinnamon AI, <https://cinnamon.ai>

AI Engineer / Researcher

Jul. 2019 – Jun. 2021
Ho Chi Minh City, Vietnam

- Built the company-wide AI service template (FastAPI serving, Docker, CircleCI fan-in/fan-out pipelines, AWS) that cut project deployment setup from 2 days to 2 hours and became the inference pipeline for nearly all client projects.
- Implemented a custom dependency-resolution and automated weekly update system to keep dozens of downstream project images reproducible and current.
- Cut inference runtime 40% on an internal text-detection model at equal F1 via structured pruning and mixed-precision training.
- Improved out-of-distribution document classification F1 from 89% to 95% while reducing inference time 20%, by ensembling an ODIN-based and a metric-learning model.
- Mentored 2 junior engineers from bootcamp through to full-time roles; ran internal ML seminars and teaching material.

PROJECTS

FOSTER – Data-efficient LLM Training via Dataset Distillation

Sole engineer, University of Queensland | in progress

2025 – 2026
PyTorch, HuggingFace, Qwen3-4B

- Built a training-data compression framework (first-order bi-level optimisation, stochastic subset sampling, distributional regulariser) that synthesises a tiny training set which reproduces full-dataset model quality.

- Matched full-data performance on three large public benchmarks from fewer than 0.1% as many training sequences, and cut the distillation compute itself by $\sim 10\times$ versus prior methods.
- Verified the compressed data transfers across architectures by fine-tuning Qwen3-4B on it – 87–95% of full-data quality at under 10% of the training time.

NaCS – Noise-aware Coreset Selection for Model Training

2025 – 2026

Sole engineer, University of Queensland | [🐙chenxing1999/nacs](#)

PyTorch, LLM fine-tuning

- Built a data-selection pipeline (submodular optimisation plus progressive label correction) that picks the highest-value training subset and repairs mislabelled examples in place.
- Recovered 93–95% of full-dataset AUC from 1% of the data, cutting total training time 75%; validated on both classical models and LLM-based ones.

SHAVER – Single-shot Model Compression for On-device Inference

2024 – 2025

Sole engineer, University of Queensland | [🐙chenxing1999/shaver](#)

PyTorch, model compression

- Built a game-theoretic parameter-attribution method enabling one-shot pruning to any target memory budget without the usual prune–retrain loop.
- Achieved 90% memory reduction at competitive accuracy, letting models run under tight edge/mobile memory budgets.

Reproducible Benchmarking Harness for Compressed Models

2024 – 2025

Sole engineer, University of Queensland | [🐙chenxing1999/recsys-benchmark](#)

Python, PyTorch, Optuna

- Built a unified evaluation harness comparing a dozen model-compression methods under matched memory and compute budgets, with tuned baselines rather than reported numbers.
- Produced practical accuracy/memory/latency trade-off guidance for deployment-constrained settings.

SKILLS

Languages & ML: Python, PyTorch, CUDA, HuggingFace Transformers, LLM fine-tuning & evaluation, classical ML, model compression & quantisation.

Engineering & MLOps: Linux, FastAPI, Docker, CircleCI, AWS, Git, Optuna, Neptune.

Spoken languages: Vietnamese (Native), English (C2), Chinese (B1).

AWARDS

Australian Government Research Training Program (RTP) Scholarship

2023 – 2027

Nomination for Excellent Student of Ho Chi Minh City

2021

[Vietnamese News Link](#)

Dean's List of HCMUS Honor Program

2020

Merit-based Scholarship of HCMUS

2017 – 2021

REFEREES

Release upon contact.